End Term Pre-Project Presentation

On

Development and Characterization of Bio-Composites from Waste Biomass for Sustainable Insulation

By

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PRESENTATION OUTLINE

1. Incorporation of Suggestions
2. Introduction.
3. Problem Statement
4. Extended Literature survey
5. Modified Objectives
6. Materials
7. Experimental setup / Model setup
8. Experimental procedure /modelling procedure
9. Results and discussions.
10. Research planning on a monthly basis
11. References

1. INCORPORATION OF SUGGESTIONS

Suggestion Received	Action Taken / Answer
Objectives were not defined/too long.	Redefined into concise, specific goals (max 2 lines each).
Presentation format was paragraph-heavy.	Converted entire presentation to bullet points for better readability.
Lack of Scale-Up Plan.	Added a dedicated slide detailing the industrial manufacturing line for these panels.
Experimental Procedure clarity.	Converted procedure into a streamlined flow sheet format.
Literature Survey depth.	Extended survey to include latest 10 research papers.

2. INTRODUCTION

Global Challenge: Conventional insulators (fiberglass, polystyrene) are fossil-fuel-based and create landfill waste, while energy demand for buildings is rising.

Sustainable Solution: Invasive aquatic biomass clogs waterways but is an abundant, fibrous raw material.

Water Bodies like Dal Lake

Waste Aquatic Biomass

Our Project:

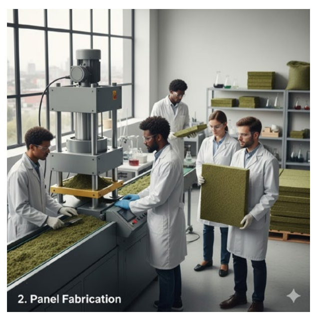
1. **Fabricate & Characterize**:
   Created bio-composite panels from local aquatic waste (water hyacinth, lily pads, lotus leaves).

2. **Optimize & Test**:
   Compared eco-friendly binders (starch, chitosan) and flame retardants (borax).

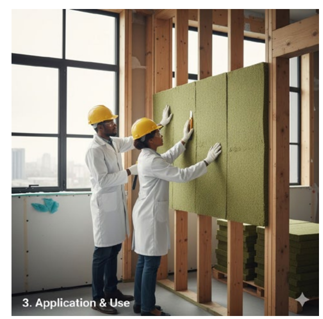
3. **Measure Performance**:
   Tested for thermal conductivity, mechanical strength, and fire safety.

4. **Goal**:
   Deliver a viable, sustainable alternative to commercial insulators.

Waste Aquatic Biomass Harvesting



Panel Fabrication



Application as insulation panels

End-of-Life & Energy Recovery (possibly)

3. PROBLEM STATEMENT

"While waste aquatic biomass offers a promising sustainable alternative to synthetic insulation, its direct application is currently hindered by inherent material deficiencies specifically **low mechanical strength**, **high flammability**, and **excessive moisture absorption:** necessitating the development of engineered bio-composites to ensure structural durability, fire safety, and long-term performance."

4. EXTENDED LITERATURE SURVEY

Author(s) & Year	Focus / Materials	Key Parameters / Conditions	Key Findings	Research Gap / Remarks
Jaktorn & Jiajitsawat (2021)	Water Hyacinth Fiber (WHF) + Natural Rubber Latex (NRL)	WHF:NRL ratios (70g WHF to 130g-170g NRL). Pressed at 100°C for 2 min.	Excellent thermal conductivity (0.0246-0.0305 W/m·K). Increasing NRL increased density.	High water absorption did not meet TISI standards. Needs significant improvement in moisture resistance.
Zhou et al. (2022)	Rice Straw + Sodium Alginate / Chitosan Binders	Immersion in $CaCl_2$ solution for crosslinking. Tested at high RH (93% & 97%) at 20°C.	Thermal conductivity 0.038-0.047 W/m·K. $CaCl_2$ crosslinking improved water resistance.	Mold growth was significant on alginate composites. Requires addition of bio-preservatives.
Pinto et al. (2021)	Corn's cob vs. Extruded Polystyrene (XPS)	SEM/EDS for microstructure. Thermography for surface temperature comparison.	Corn's cob has a similar closed-cell microstructure and chemical composition to XPS.	Lacks quantitative data on thermal conductivity. Needs development into a standardized composite panel.
Yang et al. (2020)	Review of Mycelium-Based Bio-Composites	Growth at 24-25°C, ~98% RH. Post-growth pressing increases density and strength.	Mycelium is a low-cost binder for waste. Excellent acoustic properties (>75% absorption at 1000 Hz).	Production is not standardized. Need for models to connect growth parameters to final material properties.
Felix Sahayaraj et al. (2023)	Review of Fire Retardants for Natural Fiber Composites	Methods: Cone Calorimetry, Limiting Oxygen Index (LOI), UL-94 Vertical Burning Test.	Additives (P, N compounds), coatings, and nanoparticles can significantly improve fire retardancy.	Lack of standardized testing protocols and data on long-term durability of treatments.
Kymäläinen & Sjöberg (2022)	Flax and Hemp Fibres for Insulation	Focus on fiber processing (retting, scutching) and its effect on thermal properties.	Performance is highly dependent on fiber purity and processing consistency.	Need to optimize fiber extraction to ensure consistent quality for large-scale production.
Asdrubali et al. (2021)	Review of Acoustic Properties of Natural Fibers	Sound absorption coefficient measured across various frequencies (100 Hz - 5000 Hz).	High sound absorption due to porous structure, making them multi-functional.	More research needed on acoustic performance at low frequencies and under varying humidity.
Chen et al. (2020)	Boron-Based Flame Retardants on Wood-Plastic Composites	Borax/Boric Acid added at 5-15 wt%. Flammability tested via LOI and cone calorimetry.	Boron compounds effective in increasing LOI, but can reduce mechanical strength at high conc.	Optimizing concentration to balance fire safety with mechanical and water-resistant properties.

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Aridi et al. (2023)	Date Palm Waste for Building Insulation	Composites made with polyester resin. Tested for thermal conductivity and flexural strength.	Lightweight composites with low thermal conductivity.	Poor fiber-matrix adhesion limits mechanical strength; requires fiber surface treatments.
Oushabi et al. (2022)	Composites from Sugarcane Bagasse	Alkali treatment (NaOH solution) of fibers before composite fabrication.	Alkali treatment improved mechanical strength. Good insulation properties.	High moisture absorption remains a major challenge, requiring effective hydrophobic treatments.
Wang et al. (2023)	Bio-composites from Pineapple Leaf Fiber (PALF)	PALF/Epoxy composites. Mechanical properties tested via tensile and flexural tests.	PALF has high cellulose content and mechanical strength, suitable for structural insulation.	Difficulties in large-scale PALF extraction and processing hinder commercialization.
Sair et al. (2021)	Cork-based composites	Hot pressing at 160°C. Density varied from 200-300 kg/m³.	Cork's closed-cell structure provides excellent thermal, acoustic, and fire resistance.	Limited availability and higher cost of cork compared to agricultural waste fibers.
Muthuraj et al. (2022)	Kenaf Fiber Composites	Samples exposed to accelerated weathering (UV, humidity cycles) to test durability.	Good balance of low density and high strength.	Performance degradation over time due to UV exposure needs further investigation.
Oyejobi et al. (2020)	Coconut Husk (Coir) Fiber Composites	Compression molding at 150°C. Varied fiber loading from 30-50 wt%.	Coir composites are durable but high lignin content makes them rigid and brittle.	Improving the flexibility and interfacial bonding of coir fibers within the matrix.
Zhang et al. (2024)	Review of End-of-Life Management for Bio-Composites	Pyrolysis studied at temperatures ranging from 400-600°C.	Pyrolysis is a viable method for energy recovery from end-of-life bio-composites.	Lack of data on emissions from mixed composites containing various additives.
Binici et al. (2023)	Insulation from Cotton and Textile Waste	Non-woven panels bonded with phenolic resin. Tested for thermal and acoustic properties.	Effective insulation with good acoustic properties.	Heterogeneity of waste streams makes it difficult to produce panels with consistent properties.

Author(s) & Year	Focus / Materials	Key Parameters / Conditions	Key Findings	Research Gap / Remarks
Li et al. (2021)	Hygrothermal Behavior of Bamboo Fiber Composites	Adsorption-desorption isotherms measured at different RH levels (11-97%).	Good thermal performance and moisture buffering capacity, regulating indoor humidity.	Susceptibility to insect and fungal attack requires non-toxic preservative treatments.
Kumar & Singh (2022)	Review on Potential of Natural Fibers for Insulation	General review of various fibers (jute, hemp, coir, etc.).	Confirms the vast potential of diverse natural fibers for building insulation.	A universal, standardized testing protocol for comparing different bio-materials is needed.
Liu et al. (2024)	Review of Advanced Thermal Insulation Materials from Aerogels	Freeze-drying and supercritical drying methods to create aerogels.	Cellulose aerogels offer super-insulating properties ($\lambda < 0.020$ W/m·K).	Need for cost-effective and scalable production methods to make bio-aerogels viable.
Trabelsi et al. (2020)	Hygrothermal characterization of hemp-shiv concrete	Curing time of 28 days. Tested under different temperature (23°C) and RH (50-75%) gradients.	Excellent moisture buffering capacity (MBV).	Lower compressive strength compared to traditional concrete, limiting structural applications.
Pawlowski et al. (2025)	Bio-based thermal insulation composites	Systematic literature review (2020–2025) on natural fiber and resin optimization	Summarized advances in bio-based fibers, eco-resins, and insulation applications	Need for comparative LCA among fiber and binder systems
Ahmad et al. (2022)	Natural fibre bio composites for construction	Life cycle assessment and sustainability analysis	Identified circular economy potential and recyclability improvements	Limited field data on long-term performance
Raja et al. (2023)	Bio-based insulation for energy-efficient buildings	Review of insulation λ (thermal conductivity) and fire retardancy studies	High insulation potential and fire resistance of treated natural fibers	Hybrid composite performance data lacking
Bourbia et al. (2023)	Bio-based building materials (gypsum + date palm waste)	Review of thermal and structural performance	Hybrid composites provide both structural and insulating benefits	Processing optimization needed for scalability
Cosentino et al. (2023)	Natural bio-based insulation (hemp, cork, kenaf, coir)	Experimental comparison under varying RH and density	Found superior thermal and acoustic properties for hemp and cork	Need standardized benchmarking under humidity variation

Author(s) & Year	Focus / Materials	Key Parameters / Conditions	Key Findings	Research Gap / Remarks
Zwawi (2021)	Surface modification of natural fiber bio-composites	Review of chemical and physical surface treatments	Surface modification improves adhesion and mechanical stability	Tradeoff between environmental impact and durability
Haboubi (2024)	Bio-composite insulation for cold climates	Field and lab assessment of mechanical/thermal performance	Demonstrated mechanical reliability under low temperatures	Requires validation across varying climatic zones
Curto et al. (2020)	Acoustic and thermal performance of bio-composites	Thermal and acoustic impedance testing	Demonstrated dual-function insulation properties	Relationship between density and performance not optimized
Bharath & Basavarajappa (2016)	Natural fiber biocomposite applications	Comparative review of coir, jute, and sisal composites	Good durability under acid/sulfate exposure	Outdated testing; lacks modern fire and LCA studies
Yazdi et al. (2014)	Bio-composite potential in sustainable construction	Review of natural fiber use in building materials	Early recognition of moisture control and thermal regulation	No data on hybrid or nanofiller systems

5. MODIFIED OBJECTIVES

The primary goals of this research are:

1) Develop eco-friendly bio-composite insulation panels using waste aquatic biomass.
2) Optimize binder and additive systems and quantify effects on thermal, mechanical, and flame retardant performance
3) To fabricate and characterize variants to determine the optimal formulation with acceptable mechanical strength and thermal conductivity.
4) Evaluate moisture resistance and hydrophobic strategies and measure water absorption and hygrothermal coupling
5) Assess end-of-life options including pyrolysis/energy recovery to estimate environmental benefits.

6. MATERIALS

Category	Materials Selected	Purpose / Rationale
Primary Biomass	Water Hyacinth (WH) biomass (stems + roots), Water Lily, Lotus Leaves and stems	Main structural and insulative component, utilizing abundant local waste.
Natural Binders	Corn Starch, Chitosan, Natural Rubber Latex	Eco-friendly agents to bind fibers together, offering biodegradable matrix options.
Synthetic Binders	*Optional:* Low-viscosity epoxy resin, phenolic resin	To be used in small percentages for comparison, evaluating performance enhancements.
Flame Retardants	Borax, Boric Acid, Ammonium Polyphosphate (APP)	To improve fire safety, a critical property for building materials. Eco-friendly options are prioritized.
Fillers	*Optional:* Rice Husk Ash (RHA), Nano/Micro Clay, Fly Ash	To potentially enhance mechanical strength, reduce density, and lower overall material cost.
Hydrophobic Coating	Beeswax, Linseed Oil, Silicone Spray	To improve moisture resistance, a key challenge for bio-based materials, and for performance comparison.
Solvent	Distilled Water	Used to create slurries for natural binders, ensuring a consistent mixture.

Table 1: Materials Required and its purpose

7. EXPERIMENTAL SETUP

Our research utilizes standard laboratory equipment for both fabrication and characterization of the bio-composite samples.

1) Fabrication Equipment:

- **Hot Press**: For curing and compressing the composite mixture into panels of uniform density and thickness.

Hot Press

- **Laboratory Oven**: For drying raw fibres and post-curing the samples. ASTM **D4442**

Laboratory Oven

- **Blender/Grinder**: For processing the dried water hyacinth into uniform fibres.

Grinder

- **Sieves and Sieve Shaker:** A set of standard mesh sieves will be used to classify the ground biomass by particle size. **ASTM E11**

Sieve Shaker

- **Heated Mixing Vessel / Hot Plate with Magnetic Stirrer :** Used for cooking and gelatinizing starch, melting beeswax, and homogenizing binder ingredients. **ASTM E300**

Hot Plate with Magnetic Stirrer

- **Moulds (Steel/Wooden) :** Required for shaping composite mixtures into standardized dimensions for testing. **ASTM D695**

Moulds

2) Characterization Equipment:

- **Thermal Conductivity Analyzer:** To measure the insulation capability (k-value). **ASTM C177**

Thermal Conductivity Analyzer

- **Universal Testing Machine (UTM):** To determine mechanical properties like compressive and flexural strength. **ASTM D695**

Universal Testing Machine (UTM)

- **Flammability Chamber**: To conduct flammability tests. **ASTM D3801**

- **Acoustic Impedance Tube**: To measure sound absorption coefficient. **ASTM E1050**

Flammability Chamber

Acoustic Impedance Tube

8. EXPERIMENTAL PROCEDURE

1. Collect, Wash, Sun-Dry, Grind, Sieve, Store

→

2. Binder Prep → Mix (Ratios) → Homogenize → Degas (Optional)

→

3. Pour into Moulds → Press/Hot-Press → Oven Cure → Demould → Condition

→

4. Test: Thermal, Mechanical, Water Absorption, Flammability, Acoustics

↓

5. Analyze Results → Refine Formulations

9. RESULTS AND DISCUSSION

Methods — what's done and planned :-

1) Completed:

Extensive literature review; collection, washing, sun-drying, grinding of raw biomass.

***Figure**:- Raw material collected (Water hyacinth and Water Lily)*

***Figure**:- Raw material after grinding*

Planned experimental procedure (summary):

1) **Sieve and characterize** grinded biomass (particle size distribution, bulk density).
2) **Binder preparation:** Gelatinize corn starch in water (control solids), add glycerol to 5–15 wt% relative to starch, disperse beeswax emulsion / melt-coat where required.
3) **Formulation matrix:** prepare variants (biomass: binder ratios e.g., 70:30, 60:40); include variants with $Al(OH)_3$ (5–20 wt% of composite) and borax at low concentrations for flame retardancy/biostability.
4) **Forming & curing:** fill moulds, apply pressure (hot press if available), oven-dry/condition.
5) **Characterize samples:** using thermal, physical, mechanical, water absorption, flammability, and acoustic tests with ≥3 replicates per formulation.

Scale-Up Strategy (Industrial Production)

Proposed Manufacturing Line:

1) **Mechanical Harvesting:** Amphibious harvesters for large-scale biomass collection from Water Bodies.

2) **Industrial Drying:** Rotary drum dryers to reduce moisture content to <10% efficiently.

3) **Refining:** Hammer mills or disc refiners to process tons of biomass into fibrous pulp.

4) **Resination:** Continuous spray-line to apply bio-binders (starch/latex) and fire retardants uniformly.

5) **Continuous Pressing:** Multi-daylight hot press system for mass-producing ft panels.

6) **Finishing:** Automated trimming, sanding, and hydrophobic coating application.

10. RESEARCH PLANNING ON A MONTHLY BASIS

Task	Aug & Sep	Oct	Nov & Dec	Jan & Feb	March	April
Literature Survey & Objective Finalization	✅					
Raw Material Collection & Pre-treatment		✅				
Fabrication of Preliminary Composite Batches		✅				
Characterization & Testing of Batches			✅			
Analysis & Optimization of Formulations			✅			
Fabrication of Optimized Composites				✅		
Final Comprehensive Testing				✅		
Data Compilation & In-depth Analysis					✅	

Task	Aug & Sep	Oct	Nov & Dec	Jan & Feb	March	April
Thesis Writing & Report Preparation					✓	✓
Final Presentation Preparation & Submission						✓

11. REFERENCES

1) Jaktorn, C., & Jiajitsawat, S. (2021). Production of Thermal Insulator from Water Hyacinth Fiber and Natural Rubber Latex. *Journal of Ecological Engineering, 22*(7), 134-141.

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3) Sahayaraj, A. F., et al. (2023). Fire Retardant Potential of Natural Fiber Reinforced Polymer Composites: a Review. *Journal of Natural Fibers, 20*(1).

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7) Chen, L., Li, Y., & Zhou, X. (2020).